

PATENT LANDSCAPE FOR HIGH-ACRYLONITRILE NBR SYNTHESIS AND CONTROLLED EMULSION/ORGANIC-SOLVENT POLYMERIZATION

1. ABSTRACT

This report synthesizes two primary patents relevant to high-acrylonitrile nitrile butadiene rubber (NBR) synthesis and process control. The target material is a butadiene–acrylonitrile copolymer enriched in acrylonitrile, typically requiring careful monomer feed control, polymerization temperature management, and molecular-weight regulation to achieve the desired high-ACN grade while maintaining processability. EP2478021B1 focuses directly on nitrile rubber production in organic solvents using free-radical polymerization of a conjugated diene, an α,β -unsaturated nitrile, and optional comonomers in the presence of a modifier; the patent emphasizes solution-phase synthesis, RAFT-like control chemistry, and examples using acrylonitrile/butadiene feeds at 37/63 phm under 3 bar and 100°C. It also discloses terpolymer variants incorporating hydroxyethyl acrylate or HDODA and post-polymerization isolation by ethanol precipitation. US7592409B2 is not an NBR manufacturing patent per se, but it is highly relevant because it discloses controlled emulsion polymerization technology that explicitly includes block polymers containing acrylonitrile/methacrylonitrile and conjugated diolefin segments, including acrylonitrile/butadiene-based B blocks. The patent's core contribution is in-situ emulsifier formation from latent surfactant/surfactant activator pairs, enabling controlled emulsion polymerization with conventional aqueous media, rapid initiation, and stable latex formation. Across the two patents, the main process trends are: solution polymerization in organic solvent with modifier-mediated control for direct NBR synthesis, and controlled emulsion polymerization with in-situ emulsification for architecturally complex acrylonitrile-containing block copolymers. The evidence supports high-ACN NBR synthesis strategies centered on monomer ratio selection, oxygen exclusion, controlled radical initiation, and solvent/emulsion process design rather than hydrogenation or post-modification routes.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2478021B1
- Patent Title: Nitrile rubbers and production of same in organic solvents
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses free-radical preparation of nitrile rubbers in organic solvent, using at least one conjugated diene and at least one α,β -unsaturated nitrile in the presence of a modifier. The examples show solution polymerization in dimethylacetamide with acrylonitrile and 1,3-butadiene, followed by precipitation in ethanol and vacuum drying; some examples also introduce termonomers such as hydroxyethyl acrylate or HDODA. - Vazo® 88: 284.9 mg — Example 1 - Vazo® 88: 1.17 mmol — Example 1 - Vazo® 88: 0.31 phm — Example 1 - DoPAT: 204 mg — Example 1 - DoPAT: 0.583 mmol — Example 1 - DoPAT: 0.22 phm — Example 1 - Dimethylacetamide: 50 ml — Example 1 - Dimethylacetamide: 51 phm — Example 1 - Acrylonitrile: 43 ml — Example 1 - Acrylonitrile: 653.3 mmol — Example 1 - Acrylonitrile: 37 phm — Example 1 - 1,3-Butadiene: 90 ml — Example 1 - 1,3-Butadiene: 1077 mmol — Example 1 - 1,3-Butadiene: 63 phm — Example 1 - Pressure: 3 bar — Example 1 - Reaction temperature: 100°C — Example 1 - Polymerization time: 9 hours — Example 1 - Hydroxyethyl acrylate (HEA): added to solvent prior to argon degassing — Example 6 - Reaction time: 6 hours — Example 6 - Final conversion: 39% — Example 6 - Regulator amount: 2.22 phm — Example 7 - Reaction time: 9 h — Example 7 - Final conversion: 48% — Example 7 - Styrene: 2.79 g — Example 8 - Styrene: 26.8 mmol — Example 8 - Styrene: 50 phm — Example 8 - Temperature: 100°C — Example 8 - Reaction time: 13 hours — Example 8 - Final conversion: 9.2% — Example 8 - Vazo® 88: amount shown in Table 7 — Examples 17-23 - DoPAT: amount shown in Table 7 — Examples 17-23 - DMAc: volume shown in Table 7 — Examples 17-23 - Acrylonitrile: amount shown in Table 7 — Examples 17-23 - HDODA: added in Examples 17-21 — Examples 17-23 - HEA: additionally added in Example 21 — Examples 17-23 - 1,3-Butadiene: amount shown in Table 7 — Examples 17-23 - Pressure: 3 bar — Examples 17-23 - Reaction temperature: 100°C — Examples 17-23 - Polymerization time: 9 hours — Examples 17-23 - Vazo® 88 post-dose: 128 mg — Examples 18, 20, 23 - Vazo® 88 post-dose: 0.523 mmol — Examples 18, 20, 23 - Vazo® 88 post-dose: 0.15 phm — Examples 18, 20, 23 - Post-dose solvent: 5 ml dimethylacetamide — Examples 18, 20, 23

Relevant Experimental Evidence - Example 1: Direct solution polymerization of acrylonitrile/butadiene in DMAc at 100°C and 3 bar produced NBR using Vazo® 88 and DoPAT. - Example 6: A terpolymer containing hydroxyethyl acrylate was prepared by the same solution route and stopped after 6 hours at 39% conversion. - Example 7: Lower regulator loading (2.22 phm) was used to deliberately obtain low-molecular-weight NBR, reaching 48% conversion after 9

hours. - Example 8: The low-Mn NBR from Example 7 was chain-extended with styrene to form an NBR-block-styrene terpolymer at 100°C. - Example 17: Acrylonitrile/butadiene polymerization was repeated with additional termonomers (HDODA and/or HEA) under the same 3 bar, 100°C solution conditions.

Technical Relevance This patent is directly relevant because it discloses organic-solvent synthesis of nitrile rubbers from acrylonitrile and butadiene, with explicit control over monomer feed, regulator level, and temperature. The solution process is especially pertinent for high-acrylonitrile grades where composition control and controlled molecular weight are critical.

Patent 2

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses controlled emulsion polymerization using an in-situ formed emulsifier generated from a latent surfactant and a surfactant activator in aqueous medium. Although many examples are styrene-based, the claims and abstract explicitly extend the method to block polymers containing acrylonitrile or methacrylonitrile with conjugated diolefin monomers such as isoprene and butadiene, enabling A-B-A triblock architectures. - Latent surfactant: ~0.2 grams — Example 1 - Styrene: 5 milliliters — Example 1 - Surfactant activator water phase: 5 milliliters — Example 1 - Styrene: 1,000 grams — Example 2 - Oleic acid: 60.0 grams — Example 2 - Dibenzyltrithiocarbonate: 7.2 grams — Example 2 - RO water: 4,000 grams — Example 2 - Potassium persulfate: 40.0 grams — Example 2 - Tripotassium phosphate: 40.0 grams — Example 2 - Potassium hydroxide: 16.4 grams — Example 2 - Reaction temperature: 65° C — Example 2 - Complete conversion time: less than 1.5 hours — Example 2 - Styrene: 1,000 grams — Example 3 - Hexadecyl half ester of maleic acid: 60.0 grams — Example 3 - Dibenzyltrithiocarbonate: 7.2 grams — Example 3 - RO water: 4,000 grams — Example 3 - Potassium persulfate: 40.0 grams — Example 3 - Tripotassium phosphate: 40.0 grams — Example 3 - Potassium hydroxide: 16.4 grams — Example 3 - Reaction temperature: 65° C — Example 3 - Complete conversion time: less than 3 hours — Example 3 - 0.6 phr K/Oleate soap solution: added prior to isolation — Example 3 - RO water: 200 — Example 4 - RO water: 241 — Example 5 - RO water: 197.1 — Example 5 - RO water: 84.7 — Example 5 - Styrene: 1,000 grams — Example 6 - Oleic acid: 60.0 grams — Example 6 - PBN: 10.0 grams — Example 6 - RO water: 4,000 grams — Example 6 - Potassium persulfate: 40.0 grams — Example 6 - Tripotassium phosphate: 40.0 grams — Example 6 - Potassium hydroxide: 16.4 grams — Example 6 - Reaction temperature: 75° C — Example 6 - Complete conversion time: less than 2 hours — Example 6 - Styrene: 50 grams (0.48 moles) — Example 8 - Oleic acid: 3.0 grams

(0.0106 moles) — Example 8 - PBN: 0.44 grams (0.002485 moles) — Example 8 - Distilled water: 200 grams — Example 8 - Potassium persulfate: 2.06 grams (0.00762 moles) — Example 8 - Tripotassium phosphate: 2.13 grams (0.01 moles) — Example 8 - Potassium hydroxide: 0.9 grams (0.014 moles) — Example 8 - Reaction temperature: 65° C — Example 8 - Complete conversion time: about 7 hours — Example 8 - Styrene: 200 grams (1.923 moles) — Example 10 - Oleic acid: 12.0 grams (0.0424 moles) — Example 10 - PBN: 2.0 grams (0.0113 moles) — Example 10 - Distilled water: 500 grams — Example 10 - Potassium persulfate: 8.0 grams (0.296 moles) — Example 10 - Tripotassium phosphate: 8.0 grams (0.0377 moles) — Example 10 - Potassium hydroxide: 3.6 grams (0.056 moles) — Example 10 - Reaction temperature: 75° C — Example 10 - Complete conversion time: about 3.5 hours — Example 10 - Styrene: 50.0 grams (0.48 moles) — Example 11 - PIN: equimolar amount — Example 11 - Complete conversion time: about 7 hours — Example 11 - Styrene: 100 grams of about 22% solids latex — Example 12 - Potassium persulfate: 0.5 grams (0.00184 moles) — Example 12 - Distilled water: 100 milliliters — Example 12 - n-Butyl acrylate: 22.0 grams (0.171 moles) — Example 12 - Reaction temperature: about 65° C — Example 12 - Weak exotherm: about 70° C — Example 12 - Complete conversion time: about 2 hours — Example 12 - Styrene: 50.0 grams (0.48 moles) — Example 13 - Benzaldehyde: 0.31 grams (0.00292 moles) — Example 13 - N-isopropylhydroxylamine: 1.4 grams of a 15% aqueous solution — Example 13 - Room temperature: 1 hour — Example 13 - Distilled water: 200 grams — Example 13 - Potassium persulfate: 2.06 grams (0.00762 moles) — Example 13 - Tripotassium phosphate: 2.13 grams (0.01 moles) — Example 13 - Potassium hydroxide: 0.9 grams (0.014 moles) — Example 13 - Reaction temperature: 75° C — Example 13 - Complete conversion time: about 4 hours — Example 13 - Styrene: 50.0 grams (0.48 moles) — Example 14 - Dodecylbenzenesulfonic acid: 3.0 grams (0.0092 moles) — Example 14 - Water: 1 milliliter — Example 14 - Sodium nitrite: 0.171 grams (0.002485 moles) — Example 14 - Distilled water: 200 grams — Example 14 - Potassium persulfate: 2.06 grams (0.00762 moles) — Example 14 - Tripotassium phosphate: 2.13 grams (0.01 moles) — Example 14 - Potassium hydroxide: 0.74 grams (0.0115 moles) — Example 14 - Reaction temperature: 75° C — Example 14 - Complete conversion time: about 3 hours — Example 14 - Styrene: 50 grams (0.48 moles) — Example 15 - Oleic acid: 3.0 grams (0.0106 moles) — Example 15 - 1,1-Diphenylethylene (DPE): 0.448 grams (0.002485 moles) — Example 15 - Distilled water: 200 grams — Example 15 - Potassium persulfate: 2.06 grams (0.00762 moles) — Example 15 - Tripotassium phosphate: 2.13 grams (0.01 moles) — Example 15 - Potassium hydroxide: 0.92 grams (0.0143 moles) — Example 15 - Reaction temperature: 75° C — Example 15 - Complete conversion time: about 2 hours — Example 15 - Styrene: 2,240 grams — Example 16 - Oleic acid: 134.4 grams — Example 16 - Dibenzyltrithiocarbonate: 16.0 grams — Example 16 - RO water: 3,808 grams — Example 16 - Potassium persulfate: 13.44 grams — Example 16 - Tripotassium phosphate: 89.6 grams — Example 16 - Potassium hydroxide: 36.96 grams — Example 16 - Reaction temperature: 65° C — Example 16 - Complete conversion time: less than 3 hours — Example 16 - Latex solids: 38.9% — Example 16 - Sodium nitrite: 0.69 grams (0.01 moles) — Example 17 - Oleic acid: 3.0 grams (0.0106 moles) — Example 17 - Styrene: 50 grams (0.48 moles) — Example 17 - Acetic acid: 1.0 grams (0.0167 moles) — Example 17 - Water: 2 grams — Example 17 - Distilled water: 160 grams — Example 17 - Potassium persulfate: 2.0 grams (0.00739 moles) — Example 17 - Tripotassium phosphate: 2.0 grams (0.01 moles) — Example 17 - Potassium hydroxide: 2.35 grams (0.0367 moles) — Example 17 - Reaction temperature: 75° C — Example 17 - Complete conversion time:

about 2.5 hours — Example 17 - Latex solids: 26.1% — Example 17 - Styrene: 50 grams (0.48 moles) — Example 18 - Oleic acid: 3.0 grams (0.0106 moles) — Example 18 - Crude 1-chloro-1,1-diphenylethane: approximately 0.01 moles — Example 18 - Distilled water: 200 grams — Example 18 - Potassium persulfate: 2.0 grams (0.0074 moles) — Example 18 - Tripotassium phosphate: 2.0 grams (0.0094 moles) — Example 18 - Potassium hydroxide: 1.55 grams (0.0242 moles) — Example 18 - Reaction temperature: 75° C — Example 18 - Complete conversion time: about 4 hours — Example 18 - Latex solids: 22.7% — Example 18 - Styrene: 1,300 grams (12.5 moles) — Example 19 - Oleic acid: 78.0 grams (0.2756 moles) — Example 19 - Crude 1-chloro-1,1-diphenylethane: 39.0 grams — Example 19 - Deionized RO water: 2,470 grams — Example 19 - Potassium persulfate: 50.7 grams (0.1875 moles) — Example 19 - Tripotassium phosphate: 50.7 grams (0.2388 moles) — Example 19 - Potassium hydroxide: 29.5 grams (0.602 moles) — Example 19 - Reaction temperature: 75° C — Example 19 - Complete conversion time: about 5.5 hours — Example 19 - Latex solids: 38.6% — Example 19 - 1-hexene: 25 grams — Example 20 - n-Butyl acrylate: 25 grams — Example 20 - Dipyrityl: 0.38 grams — Example 20 - Oleic acid: 4.0 grams — Example 20 - Ethyl 2-bromoisobutyrate: 0.47 grams — Example 20 - 85% KOH: 1.7 grams — Example 20 - Distilled water: 139 milliliters — Example 20 - Copper sulfate pentahydrate solution: 60.9 grams of a 1% aqueous solution — Example 20 - Hydrazine hydrate: 2 drops — Example 20 - Reaction temperature: about 67° C — Example 20 - Reaction time: 1.5 hours — Example 20 - Latex solids: 10.9% — Example 20 - 1-hexene incorporation: 5.5 mole % — Example 20

Relevant Experimental Evidence - Example 1: Demonstrated in-situ emulsification from latent surfactant/activator pairs in styrene/water systems. - Example 2: Showed RAFT-controlled emulsion polymerization at 65°C with complete conversion in less than 1.5 hours. - Example 3: Showed in-situ generated polymerizable surfactant enabling rapid conversion and stable latex formation. - Example 12: Demonstrated seeded emulsion chain extension to a block copolymer with n-butyl acrylate at about 65°C. - Example 20: Demonstrated ATRP emulsion copolymerization with 1-hexene and n-butyl acrylate, confirming broader controlled-emulsion scope.

Technical Relevance This patent is relevant because its claims explicitly cover block polymers containing acrylonitrile or methacrylonitrile with conjugated diolefins such as butadiene or isoprene. Its in-situ emulsifier strategy is important for controlled aqueous synthesis routes that could be adapted to high-acrylonitrile NBR-related block architectures and latex processing.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2478021B1 is the more direct high-ACN NBR synthesis disclosure: it uses acrylonitrile/butadiene solution polymerization in organic solvent, with explicit monomer feed levels of 37 phm ACN and 63 phm butadiene in Example 1.
- US7592409B2 is broader and more platform-oriented: it does not exemplify NBR synthesis directly, but its claims explicitly include acrylonitrile/methacrylonitrile plus conjugated diolefin

blocks, making it relevant to NBR-based block copolymers rather than conventional random NBR.

- EP2478021B1 uses organic-solvent free-radical polymerization with modifier/RAFT-like control chemistry (DoPAT) and initiator Vazo® 88, whereas US7592409B2 centers on controlled emulsion polymerization with in-situ emulsifier formation and controlled radical methods such as RAFT, NMP, ATRP, and DT.
- EP2478021B1 operates at 100°C and 3 bar with polymerization times of 6-9 hours in the examples, while US7592409B2 commonly uses 65-75°C and reaches complete conversion in roughly 1.5-7 hours depending on the control system.
- EP2478021B1 isolates polymer by ethanol precipitation and vacuum drying after solvent polymerization; US7592409B2 generally forms stable latexes and uses stripping/coagulation only in selected examples.
- For high-acrylonitrile NBR, EP2478021B1 provides the clearest evidence for direct composition control, while US7592409B2 provides a useful emulsion-processing framework for acrylonitrile-containing block copolymers and latex-based manufacturing.

4. CONCLUSION

The evidence indicates two complementary synthesis directions for high-acrylonitrile NBR-related materials. The most directly relevant route is EP2478021B1, which discloses solution polymerization of acrylonitrile and 1,3-butadiene in DMAc with controlled radical chemistry, oxygen exclusion, 3 bar butadiene feed, and 100°C initiation; this is the strongest patent evidence for direct NBR manufacture with tunable acrylonitrile content. US7592409B2 contributes a broader controlled-emulsion platform that explicitly encompasses acrylonitrile/diolefin block polymers and demonstrates rapid, stable latex formation via in-situ emulsifier generation. Based on the evidence, the recommended synthesis parameters for high-ACN NBR are: acrylonitrile-rich monomer feed control, rigorous oxygen removal, controlled radical initiation, and either organic-solvent solution polymerization for direct NBR or controlled emulsion polymerization when block architecture or latex processing is desired.

5. REFERENCES

- EP2478021B1 | Nitrile rubbers and production of same in organic solvents | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2478021B1/en>
- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>